

Moments and Singular Perturbation

Yoonjae Hwang

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1 Problem Setup

SYSTEM:

$$\dot{x} = A_{11}x + A_{12}z + B_1u \quad (1)$$

$$\varepsilon \dot{z} = A_{21}x + A_{22}z + B_2u \quad (2)$$

$$y = C_1x \quad (3)$$

No z term as otherwise we would have feedforward in the reduced model.

In state space form:

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ \frac{1}{\varepsilon}B_2 \end{bmatrix} u \quad (4)$$

SIGNAL:

$$\dot{w}_1 = S_{11}w_1 + S_{12}w_2 \quad (5)$$

$$\varepsilon \dot{w}_2 = S_{21}w_1 + S_{22}w_2 \quad (6)$$

REDUCED MODEL SIGNAL:

$$S_{22}w_2 + S_{21}w_1 = 0 \quad (7)$$

$$w_2 = -S_{22}^{-1}S_{21}w_1 \quad (8)$$

$$\dot{w}_1 = (S_{11} - S_{12}S_{22}^{-1}S_{21})w_1 \quad (9)$$

$$u = (L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \quad (10)$$

REDUCED MODEL OF THE SYSTEM:

$$A_{22}z + A_{21}x + B_2u = 0 \quad (11)$$

$$z = -A_{22}^{-1}(A_{21}x + B_2u) \quad (12)$$

$$\dot{x} = (A_{11} - A_{12}A_{22}^{-1}A_{21})x + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \quad (13)$$

FORWARD MOMENTS

$$M_g := [C_1\Pi_{x1} \quad C_1\Pi_{x2}] \quad (14)$$

FORWARD SYLVESTER EQUATION FOR THE FULL SYSTEM:

$$\begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1L_{g1} & B_1L_{g2} \\ \frac{1}{\varepsilon}B_2L_{g1} & \frac{1}{\varepsilon}B_2L_{g2} \end{bmatrix} = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \frac{1}{\varepsilon}S_{21} & \frac{1}{\varepsilon}S_{22} \end{bmatrix} \quad (15)$$

FORWARD SYLVESTER EQUATION FOR THE REDUCED MODEL

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\tilde{\Pi} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) = \tilde{\Pi}(S_{11} - S_{12}S_{22}^{-1}S_{21}) \quad (16)$$

Can we show that as $\varepsilon \rightarrow 0$, the forward sylvester equation for the full system tends towards the forward sylvester equation for the reduced model?

2 Proof (I guess?)

Expanding out (14) we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \Pi_{x1}S_{11} + \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (17a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12} + \frac{1}{\varepsilon}\Pi_{x2}S_{22}, \quad (17b)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \Pi_{z1}S_{11} + \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \quad (17c)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12} + \frac{1}{\varepsilon}\Pi_{z2}S_{22}. \quad (17d)$$

Multiplying (16c) and (16d) by ε we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \Pi_{x1}S_{11} + \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (18a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12} + \frac{1}{\varepsilon}\Pi_{x2}S_{22}, \quad (18b)$$

$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2L_{g1} = \varepsilon\Pi_{z1}S_{11} + \Pi_{z2}S_{21}, \quad (18c)$$

$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2L_{g2} = \varepsilon\Pi_{z1}S_{12} + \Pi_{z2}S_{22}. \quad (18d)$$

We now consider the ε series expansion of Π_{x1} , Π_{x2} , Π_{z1} , Π_{z2} . We can see that by multiplying (17a) and (17b) by ε , and setting ε to 0 we have that the zeroth order term for Π_{x2} is 0.

We then write:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}S_{11} + \Pi_{x2}^{(1)}S_{21} + O(\varepsilon), \quad (19a)$$

$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12} + \Pi_{x2}^{(1)}S_{22} + O(\varepsilon), \quad (19b)$$

$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21} + O(\varepsilon), \quad (19c)$$

$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = \Pi_{z2}^{(0)}S_{22} + O(\varepsilon) \quad (19d)$$

Each of the 4 equations in (18) are essentially polynomials in epsilon.

After equating coefficients, we can write $\Pi_{x2}^{(1)}$ as such (Assuming S_{22} is invertible):

$$\Pi_{x2}^{(1)} = \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} - \Pi_{x1}^{(0)}S_{12} \right) S_{22}^{-1} \quad (20)$$

Substituting (19) into (18a) we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}S_{11} + \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} - \Pi_{x1}^{(0)}S_{12} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (21)$$

Rearranging we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (22)$$

Similarly, from (18c) we can substitute in an expression for $\Pi_{z1}^{(0)}$:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}A_{22}^{-1} \left(\Pi_{z2}^{(0)}S_{21} - A_{21}\Pi_{x1}^{(0)} - B_2L_{g1} \right) + B_1L_{g1} = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (23)$$

Rearranging we have:

$$\begin{aligned} & (A_{11} - A_{12}A_{22}^{-1}A_{21})\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)L_{g1} \\ & + \left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1} \right) S_{21} \\ & = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + O(\varepsilon) \end{aligned} \quad (24)$$

Simplifying the middle (...) S_{21} term using (18d):

$$\left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - A_{12}\Pi_{z2}^{(0)}S_{22}^{-1} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (25)$$

$$= \left(A_{12}A_{22}^{-1} \left(\Pi_{z2}^{(0)}S_{22} - A_{22}\Pi_{z2}^{(0)} \right) S_{22} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (26)$$

$$= \left(A_{12}A_{22}^{-1}B_2L_{g2}S_{22}^{-1} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (27)$$

$$= - \left(B_1 - A_{12}A_{22}^{-1}B_2 \right) L_{g2}S_{22}^{-1}S_{21} \quad (28)$$

Substituting (27) back into (23) and factorising we have:

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) \quad (29)$$

$$= \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + O(\varepsilon) \quad (30)$$

which is of $O(\varepsilon)$ error from (15). Thus as $\varepsilon \rightarrow 0$, (29) approaches (15), and the matrix $\Pi_{x1} \rightarrow \Pi_{x1}^{(0)} = \tilde{\Pi}$.

3 Next Steps

- Expression for Π_{x2} (Is it needed?)
- Higher order epsilon expansion?
- Assumptions, and rigour?

4 S_{11} and S_{22} are 0

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (31a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12}, \quad (31b)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \quad (31c)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12}. \quad (31d)$$

Similar approach to before, we'll have $\Pi_{x2}^{(0)} = 0$.

Considering zeroth order terms:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x2}^{(1)}S_{21}, \quad (32a)$$

$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12}, \quad (32b)$$

$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21}, \quad (32c)$$

$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = 0 \quad (32d)$$

Assuming A_{22} is invertible:

$$\Pi_{z2}^{(0)} = -A_{22}^{-1}B_2L_{g2} \quad (33)$$

Assuming S_{12} is invertible, (33) $\rightarrow$ (32b):

$$\Pi_{x1}^{(0)} = (-A_{12}A_{22}^{-1}B_2L_{g2} + B_1L_{g1})S_{12}^{-1} \quad (34)$$

And $\Pi_{z1}^{(0)}$ can be extracted from (32c). Simulation results doesnt seem to agree. Might be implementation issue.

5 Thoughts

- Canonical form - is there a coordinate change that we can exploit?
- Making a family of approximations? what to do when A_{22} is not Hurwitz?
- Realisation?
- Controllability and observability conditions
- Is there a limit for the difference? $\Pi w_g - x$

6 Descriptor Systems

One thing to note that you could represent singular perturbation systems as a descriptor system. Scarlotti wrote a paper on singular systems and moment matching, but something different is that for us, our signal is also a singular perturbation system. Another thing to note, when you consider the scenario where S_{11} and S_{22} are 0, the signal is a pure integrator, but also when you set $\varepsilon = 0$, you have $S_{21}w_1 = 0$ which is ambiguous. A reduced model for the signal is not well defined. So a better approach might be to consider finite ε and then take the limit as $\varepsilon \rightarrow 0$.

6.1 New Definition of Moments

I think a better definition for the matrix Π is to consider it as a linear transformation of the signal w , such that the difference between Πw and x is governed by the open loop system dynamics. Then, the Sylvester equation becomes a condition that Π must satisfy for it to be a valid linear transformation. This allows us to move away from the frequency domain definition of moments which relies on the system having a steady state response. Even for unstable systems, we can still compute a solution for the Sylvester equation when there is no steady state response.

$$\frac{d}{dt}(x - \Pi w) = A(x - \Pi w) \quad (35)$$

This way a nonlinear extension should be more straightforward, as Π will probably be some nonlinear operation on w .

7 Descriptor System Representation

Using a descriptor system representation:

$$\begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u \quad (36)$$

The signal is similarly:

$$\begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix} \begin{bmatrix} \dot{w}_1 \\ \dot{w}_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad (37)$$

Π is then defined as:

$$\begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix} \frac{d}{dt} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right) = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right) \quad (38)$$

The resulting Sylvester equation for the whole system then becomes:

$$A\Pi + BL_g = E_1\Pi E_2^{-1}S \quad (39)$$

where

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad E = \begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix}, \quad S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad \Pi = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix}, \quad L_g = [L_{g1} \quad L_{g2}] \quad (40)$$

Setting S_{11} and S_{22} to 0, we have:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} [L_{g1} \quad L_{g2}] = \begin{bmatrix} \Pi_{x1} & \frac{1}{\varepsilon}\Pi_{x2} \\ \varepsilon\Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} 0 & S_{12} \\ S_{21} & 0 \end{bmatrix} \quad (41)$$

The resulting 4 equations are then:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (42a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12}, \quad (42b)$$

$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2L_{g1} = \Pi_{z2}S_{21}, \quad (42c)$$

$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2L_{g2} = \varepsilon\Pi_{z1}S_{12}. \quad (42d)$$

Using expansion in ε , we can find the where the partitions of Π will converge to as $\varepsilon \rightarrow 0$. When S_{22} is 0, it is singular and the reduced model for the signal is undefined. The significance of this representation is that, even when there is no reduced model for the signal, we can still find a meaningful Π that relates the system and the signal.

7.1 remarks

Descriptor systems initial conditions?

- Defintion of descriptor sylvester equation
- solveability and proof of convergence
- system solution and relation to moment, steady-state